

A4.3.3 Hyperparameter tuning



A4.3.3 Explain the role of hyperparameter tuning when evaluating supervised learning algorithms.

A4.3.3 explains the role of hyperparameter tuning in optimising supervised learning algorithms. It details crucial evaluation metrics like accuracy, precision, recall, and the F1 score (calculated as the harmonic mean of precision and recall, used to assess models like CNNs). The topic also covers common training issues such as overfitting and underfitting. Hyperparameter tuning helps address these issues to ensure accurate model performance

Introduction

So, we've built a classification model. Congratulations! But how do we know if it's any good? Simply building a model is only half the battle. In this section, we'll explore the crucial process of evaluating a model's performance and tuning it to make it as effective as possible.

The Accuracy Paradox: Why "Accuracy" Isn't Enough

The most obvious way to measure performance is **accuracy**—the percentage of predictions the model got right. However, accuracy can be very misleading, especially with **imbalanced datasets**.

Consider this scenario: You build a model to detect a rare disease that only affects 1% of the population. A lazy, useless model that simply predicts "No Disease" for every single person will be **99% accurate**, but it will fail to identify any of the sick patients, making it completely worthless.

This is the Accuracy Paradox. We need more nuanced metrics to understand a model's true performance.

The Confusion Matrix: A Deeper Look at Performance

To get a better picture, we use a **Confusion Matrix**. It's a table that breaks down the predictions of a classification model, showing us exactly where it succeeded and where it failed.

		Predicted Class	
		Positive	Negative
Actual Class	Positive	True Positive (TP): Correctly predicted positive. (You said spam, it was spam).	False Negative (FN): Incorrectly predicted negative. (You said not spam, it was spam. A dangerous miss).
	Negative	False Positive (FP): Incorrectly predicted positive. (You said spam, it was not spam. A false alarm).	True Negative (TN): Correctly predicted negative. (You said not spam, it was not spam).

The Confusion Matrix

- **True Positive (TP):** The model correctly predicted the positive class. (e.g., It said "spam," and it *was* spam).
- **True Negative (TN):** The model correctly predicted the negative class. (e.g., It said "not spam," and it *was not* spam).
- **False Positive (FP):** The model incorrectly predicted the positive class. This is a **"false alarm."** (e.g., It said "spam," but it was an important email).
- **False Negative (FN):** The model incorrectly predicted the negative class. This is a **"dangerous miss."** (e.g., It said "not spam," but it *was* a spam email).

Key Evaluation Metrics

Using the values from the confusion matrix, we can calculate more useful metrics:

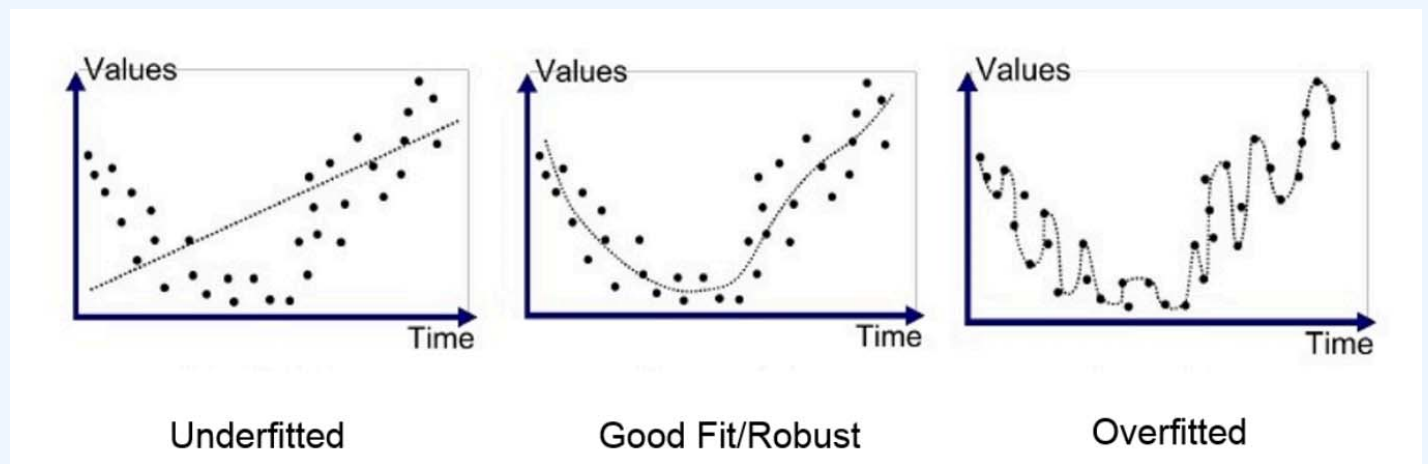
1. **Precision:** Measures how reliable the positive predictions are.
 - **Formula:** $TP / (TP + FP)$
 - **In words:** "Of all the times the model predicted 'Positive', what percentage were actually correct?"
 - **High Precision** means the model has very few false alarms.
2. **Recall (or Sensitivity):** Measures how well the model finds all the actual positives.

- **Formula:** $TP / (TP + FN)$
- **In words:** "Of all the actual 'Positives' that existed in the data, what percentage did the model find?"
- **High Recall** means the model has very few dangerous misses.

3. **F1 Score:** The **harmonic mean** of Precision and Recall. It provides a single score that balances both metrics. The F1 score will only be high if *both* precision and recall are high, making it a good general measure of a model's performance.

Overfitting and Underfitting

The goal of training is to create a model that generalises well to new, unseen data. Two common failure modes are overfitting and underfitting.



- **Underfitting:** The model is too simple to capture the underlying pattern in the data. It performs poorly on both the training data and new data.
- **Overfitting:** The model is too complex and learns the training data too well, including its random noise. It performs perfectly on the training data but fails badly on new data because it hasn't learned the general pattern.

The Role of Hyperparameter Tuning

So how do we find the "Good Fit" and avoid these problems? The answer is **hyperparameter tuning**.

- A **hyperparameter** is a configuration setting that we choose *before* the training process begins. It's like a knob or a dial on the algorithm. For example, the 'k' in the K-Nearest

Neighbours algorithm is a hyperparameter.

- **Hyperparameter Tuning** is the process of systematically trying different values for these settings to find the combination that produces the best-performing model (e.g., the one with the highest F1 score on a test dataset). This is the key to finding the right level of complexity to avoid both underfitting and overfitting.

Flashcards

Use this Quizlet to learn all the key terms in this section.



Ready to play?

Match all the terms with their definitions as fast as you can. Avoid wrong matches, they add extra time!

Start game

[View this flashcard set](#)

Choose a Study Mode 

Knowledge Check: Multiple Choice Quiz

Test your understanding of the core concepts.

1

A model is trained to detect fraudulent transactions. A "False Positive" in this context would be:

- A. ☐ A legitimate transaction is correctly identified as not fraudulent.

B. ☐ A fraudulent transaction is missed by the model.

C. ☒ A legitimate transaction is incorrectly flagged as fraudulent.

D. ☐ A fraudulent transaction is correctly identified.

2

Which metric answers the question: "Of all the actual fraudulent transactions, what percentage did our model successfully catch?"

A. ☐ Precision

B. ☐ F1 Score

C. ☒ Recall

D. ☐ Accuracy

3

A model that is too simple and fails to learn the underlying pattern in the data is said to be:

A. ☒ Underfitting

B. ☐ A "black box" model

C. ☐ A hyperparameter

D. ☐ Overfitting

4

What is the primary purpose of hyperparameter tuning?

A. ☐ To clean the training data.

- B. ☐ To choose the right type of algorithm (e.g., K-NN vs. Decision Tree).
- C. ☐ To calculate the F1 score.
- D. ☒ To find the optimal configuration settings for a model to improve its performance.

5

In a medical diagnosis model for a serious disease, which metric is often considered the most important to maximise in order to avoid dangerous misses?

- A. ☐ The number of True Negatives
- B. ☒ Recall
- C. ☐ Accuracy
- D. ☐ Precision

6

The F1 Score is the harmonic mean of which two metrics?

- A. ☐ Accuracy and Recall
- B. ☐ True Positives and True Negatives
- C. ☐ Accuracy and Precision
- D. ☒ Precision and Recall

7

A model that performs with 100% accuracy on its training data but only 50% accuracy on new data is a classic sign of:

- A. ☐ An imbalanced dataset

B. ☐ A good model

C. ☒ Overfitting

D. ☐ Underfitting

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Which of the following is an example of a hyperparameter?

A. ☐ The number of False Positives.

B. ☐ The number of data points in the training set.

C. ☒ The value of 'k' in a K-Nearest Neighbours algorithm.

D. ☐ The final accuracy of a model.

9

Which metric answers the question: "When our model predicts a transaction is fraudulent, how often is it correct?"

A. ☐ Accuracy

B. ☐ Recall

C. ☒ Precision

D. ☐ F1 Score

10

A confusion matrix is a tool used to:

A. ☐ Visualise the training data.

B. ☐ Tune hyperparameters.

C. ☐ Choose which algorithm to use.

D. ☐ Analyse the performance of a classification model.

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